PRECAST CONCRETE SPAN & LOAD TABLES

Tables for Concrete Building Project
Structural Systems 1 ARC 2713 Fall 2025

HOLLOWCORE PLANK

- 4HC6
- 4HC8

DOUBLE TEE

- 8DT12
- 8DT16
- 8DT18
- 8DT20
- 8DT24
- 10DT24

STAIRS

- Thickness & Span
- Bearing Points

BEAMS

- Rectangular
- L-Shape
- T-Shape

COLUMNS

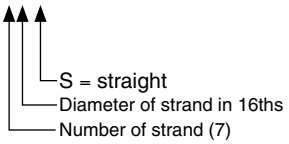
- 16 x 16
- 18 x 18
- 20 x 20
- 24 x 24
- 28 x 28
- 32 x 32

VERTICAL WALL PANELS

- Solid
- Sandwich
- Tee

3.6 Hollow-Core Load Tables

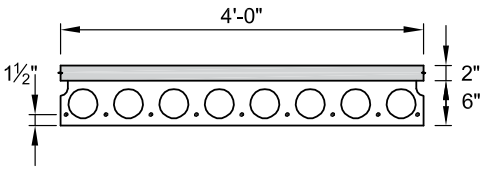
Strand Pattern Designation
76-S



Safe loads shown include dead load of 10 lb/ft² for untopped members and 15 lb/ft² for topped members. Remainder is live load. Long-time cambers include superimposed dead load but do not include live load.

Capacity of sections of other configurations are similar. For precise values, see local hollow-core manufacturer.

4'-0" × 6"
Normalweight Concrete



f'_c = 5000 psi

f_{pu} = 270,000 psi

Section Properties

		No Topping		2 in. topping	
A	=	187	in.²	-	
I	=	763	in.⁴	1640	in.⁴
y_b	=	3.00	in.	4.14	in.
y_t	=	3.00	in.	3.86	in.
S_b	=	254	in.³	396	in.³
S_t	=	254	in.³	425	in.³
wt	=	195	lb/ft	295	lb/ft
DL	=	49	lb/ft²	74	lb/ft²
V/S	=	1.73	in.		

Key
290– Safe superimposed service load, lb/ft²
0.1 – Estimated camber at erection, in.
0.2 – Estimated long-time camber, in.

4HC6

Table of safe superimposed service load, lb/ft², and cambers, in. **No Topping**

Strand designation code	Span, ft																				
	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
66-S	290	258	230	206	188	171	157	145	129	114	100	88	77	68	59	52	46	40	33	28	
	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.1	0.0	–0.1	–0.2	–0.4	–0.5	–0.7		
	0.2	0.2	0.2	0.2	0.3	0.3	0.2	0.2	0.2	0.1	0.1	0.0	–0.1	–0.3	–0.5	–0.7	–0.9	–1.2	–1.5	–1.9	
76-S		302	271	245	223	205	187	172	155	136	120	105	93	82	73	65	57	49	42	36	31
		0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.1	0.1	0.0	–0.1	–0.3	–0.4	–0.6
		0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.1	0.0	–0.1	–0.2	–0.4	–0.7	–0.9	–1.2	–1.6	–2.0
96-S		385	345	315	285	262	243	225	201	177	157	139	124	110	99	88	78	68	60	53	46
		0.3	0.3	0.3	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.4	0.3	0.3	0.1	0.0	–0.1
		0.3	0.4	0.4	0.5	0.5	0.5	0.6	0.6	0.6	0.5	0.5	0.4	0.3	0.2	0.1	–0.1	–0.3	–0.6	–0.9	–1.3
87-S		391	351	316	288	265	245	226	210	197	185	167	149	134	119	107	95	85	76	68	60
		0.3	0.4	0.4	0.5	0.5	0.6	0.6	0.7	0.7	0.7	0.7	0.8	0.8	0.7	0.7	0.7	0.6	0.5	0.4	0.3
		0.4	0.5	0.5	0.6	0.7	0.7	0.7	0.8	0.8	0.8	0.8	0.7	0.7	0.6	0.5	0.3	0.2	0.0	–0.3	–0.6
97-S		435	391	354	324	297	274	254	236	220	203	186	166	148	133	119	107	96	86	78	70
		0.4	0.4	0.5	0.5	0.6	0.7	0.7	0.8	0.8	0.9	0.9	0.9	0.9	1.0	0.9	0.9	0.9	0.8	0.7	0.6
		0.5	0.6	0.6	0.7	0.8	0.8	0.9	0.9	1.0	1.0	1.0	1.0	0.9	0.9	0.8	0.7	0.5	0.3	0.1	–0.2

4HC6 + 2

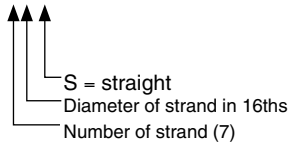
Table of safe superimposed service load, lb/ft², and cambers, in. **2 in. Normalweight Topping**

Strand designation code	Span, ft																			
	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	
66-S	333	300	272	248	227	210	182	158	136	113	93	75	59	46	34					
	0.2	0.2	0.2	0.2	0.2	02	0.2	0.2	0.2	0.2	0.1	0.1	0.0	–0.1	–0.2					
	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.0	–0.1	–0.2	–0.3	–0.5	–0.7	–0.9	–1.2					
76-S		357	324	296	272	248	216	188	163	137	115	95	78	63	50	38	27			
		0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.1	0.1	–0.0	–0.1	–0.3			
		0.2	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.0	–0.2	–0.3	–0.5	–0.7	–0.9	–1.2	–1.5			
96-S			418	384	353	319	279	245	216	186	160	137	116	98	82	68	55	43	33	
			0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.4	0.3	0.3	0.1	0.0	–0.1	
			0.4	0.4	0.4	0.4	0.4	0.4	0.3	0.3	0.2	0.1	–0.1	–0.3	–0.5	–0.7	–1.0	–1.4	–1.7	
87-S			422	388	359	332	309	288	258	224	195	169	147	127	109	94	80	67	55	
			0.5	0.5	0.6	0.6	0.7	0.7	0.7	0.7	0.8	0.8	0.7	0.7	0.7	0.6	0.5	0.4	0.3	
			0.5	0.5	0.5	0.6	0.6	0.6	0.5	0.5	0.4	0.4	0.2	0.1	–0.1	–0.3	–0.5	–0.8	–1.2	
97-S			473	433	400	371	346	323	288	251	219	192	168	146	127	110	95	82	70	
			0.5	0.6	0.7	0.7	0.8	0.8	0.9	0.9	0.9	0.9	1.0	0.9	0.9	0.9	0.8	0.7	0.6	
			0.6	0.6	0.7	0.7	0.7	0.7	0.7	0.7	0.6	0.6	0.5	0.4	0.2	0.0	–0.2	–0.5	–0.8	

Strength is based on strain compatibility; bottom tension is limited to $7.5\sqrt{f'_c}$; see pages 3–8 through 3–11 for explanation.
See item 3, note 4, Section 3.3.2 for explanation of vertical line.

3.6 Hollow-Core Load Tables (cont.)

Strand Pattern Designation
76-S

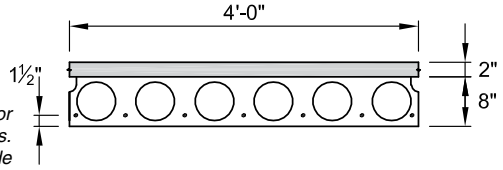


Safe loads shown include dead load of 10 lb/ft² for untopped members and 15 lb/ft² for topped members. Remainder is live load. Long-time cambers include superimposed dead load but do not include live load.

Capacity of sections of other configurations are similar. For precise values, see local hollow-core manufacturer.

Key
385– Safe superimposed service load, lb/ft²
0.1 – Estimated camber at erection, in.
0.2 – Estimated long-time camber, in.

4'-0" × 8"
Normalweight Concrete



f'_c = 5000 psi

f_{pu} = 270,000 psi

Section Properties

		No Topping		2 in. topping	
A	=	215	in.²	-	
I	=	1666	in.⁴	3071	in.⁴
y_b	=	4.00	in.	5.29	in.
y_t	=	4.00	in.	4.71	in.
S_b	=	417	in.³	581	in.³
S_t	=	417	in.³	652	in.³
wt	=	224	lb/ft	324	lb/ft
DL	=	56	lb/ft²	81	lb/ft²
V/S	=	1.92	in.		

4HC8

Table of safe superimposed service load, lb/ft², and cambers, in. **No Topping**

Strand designation code	Span, ft																														
	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
66-S	385	345	313	283	260	240	223	204	179	158	140	124	110	98	87	77	69	61	54	48	43	38	33	29						
	0.1	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.1	0.0	0.0	–0.1	–0.2	–0.3	–0.5	–0.6						
	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.2	0.1	0.0	–0.1	–0.2	–0.3	–0.5	–0.7	–0.9		–1.4						
76-S	449	407	367	334	309	285	263	242	213	188	167	149	133	119	106	95	86	77	69	62	55	50	44	39	35	31	26			
	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	0.1	0.0	–0.1	–0.2	–0.4	–0.5	–0.7	–0.9			
	0.2	0.2	0.3	0.3	0.3	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.3	0.3	0.2	0.1	0.0	–0.1	–0.2	–0.4	–0.6	–0.8	–1.1	–1.4	–1.7	–2.0			
58-S	422	380	346	316	290	267	247	231	216	202	190	179	169	160	144	130	118	107	97	88	80	72	66	60	54	48	42	37	32	28
	0.2	0.2	0.3	0.3	0.3	0.4	0.4	0.5	0.5	0.5	0.5	0.6	0.6	0.6	0.6	0.6	0.6	0.6	0.5	0.5	0.5	0.4	0.3	0.2	0.1	0.0	–0.4	–0.3	–0.5	–0.7	–0.9
	0.3	0.3	0.4	0.4	0.5	0.5	0.6	0.6	0.6	0.7	0.7	0.7	0.7	0.7	0.7	0.6	0.6	0.5	0.4	0.3	0.2	0.0	–0.2	–0.4	–0.4	–0.6	–0.9	–1.2	–1.6	–2.0	–2.4
68-S	476	430	393	361	332	309	286	269	253	235	223	209	200	180	165	153	142	132	121	110	101	92	84	77	70	63	56	51	45	40
	0.3	0.3	0.3	0.4	0.4	0.5	0.5	0.6	0.6	0.7	0.7	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.7	0.7	0.6	0.5	0.4	0.2	0.1	–0.1	–0.3	
	0.3	0.4	0.5	0.5	0.6	0.6	0.7	0.7	0.8	0.8	0.9	0.9	1.0	1.0	1.0	1.0	0.9	0.9	0.9	0.8	0.7	0.6	0.4	0.2	0.0	–0.2	–0.5	–0.8	–1.1	–1.5	
78-S	488	442	402	370	341	318	295	275	259	241	229	215	203	195	180	168	157	144	135	126	118	110	101	92	84	77	70	64	58	52
	0.3	0.3	0.4	0.5	0.5	0.6	0.6	0.7	0.7	0.8	0.9	0.9	1.0	1.0	1.0	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.0	0.9	0.8	0.7	0.6	0.5	0.3	
	0.4	0.5	0.5	0.6	0.7	0.8	0.8	0.9	1.0	1.0	1.1	1.2	1.2	1.2	1.3	1.3	1.3	1.3	1.3	1.2	1.2	1.1	1.0	0.8	0.7	0.5	0.3	0.0	–0.3	–0.7	

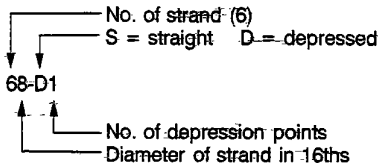
4HC8 + 2

Table of safe superimposed service load, lb/ft², and cambers, in. **2 in. Normalweight Topping**

Strand designation code	Span, ft																														
	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40			
66-S	400	365	333	308	282	256	224	197	173	153	135	119	105	93	82	68	56	45	36	26											
	0.2	0.2	0.2	0.2	0.2	0.2	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.2	0.1	0.0	–0.0	–0.1	–0.2	–0.3											
	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.1	0.1	0.0	–0.1	–0.2	–0.3	–0.4	–0.6	–0.7	–0.9	–1.2	–1.4											
76-S	474	435	396	366	340	304	267	235	208	184	164	146	130	116	103	88	74	62	51	41	31										
	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.4	0.4	0.4	0.3	0.3	0.3	0.3	0.2	0.2	0.1	–0.0	–0.1	–0.2										
	0.2	0.2	0.3	0.3	0.3	0.3	0.3	0.3	0.2	0.2	0.2	0.1	0.0	–0.1	–0.2	–0.4	–0.5	–0.7	–0.9	–1.2	–1.4										
58-S	445	405	374	342	318	298	275	260	243	228	217	196	177	159	143	126	110	95	82	70	59	49	40	32							
	0.3	0.3	0.3	0.4	0.4	0.5	0.5	0.5	0.5	0.6	0.6	0.6	0.6	0.6	0.6	0.5	0.5	0.5	0.1	0.3	0.2	0.1	0.0	–0.1							
	0.3	0.3	0.4	0.4	0.4	0.4	0.5	0.5	0.5	0.5	0.4	0.3	0.3	0.3	0.2	0.1	–0.1	–0.2	–0.4	–0.6	–0.9	–1.2	–1.5	–1.8							
68-S	463	426	393	366	342	319	299	282	267	251	239	216	195	177	158	140	124	110	97	84	73	62	53	44	36	28					
	0.4	0.4	0.5	0.5	0.6	0.6	0.7	0.7	0.7	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.8	0.7	0.7	0.6	0.5	0.4	0.2	0.1	–0.1				
	0.4	0.5	0.5	0.6	0.6	0.6	0.6	0.7	0.7	0.7	0.6	0.6	0.6	0.6	0.5	0.4	0.3	0.2	0.0	–0.2	–0.4	–0.6	–0.9	–1.2	–1.6	–2.0	–2.4				
78-S	472	435	402	375	348	325	305	288	273	257	245	232	220	207	186	167	149	133	119	106	94	83	73	64	55	46	38				
	0.5	0.5	0.6	0.6	0.7	0.7	0.8	0.9	0.9	1.0	1.0	1.0	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.0	0.9	0.9	0.7	0.6	0.5	0.3			
	0.5	0.6	0.6	0.7	0.7	0.8	0.8	0.8	0.9	0.9	0.9	0.9	0.8	0.8	0.7	0.7	0.6	0.4	0.3	0.1	–0.1	–0.3	–0.6	–0.9	–1.3	–1.7	–2.2				

Strength is based on strain compatibility; bottom tension is limited to $7.5\sqrt{f'_c}$; see pages 3–8 through 3–11 for explanation.
See item 3, note 4, Section 3.3.2 for explanation of vertical line.

Strand Pattern Designation



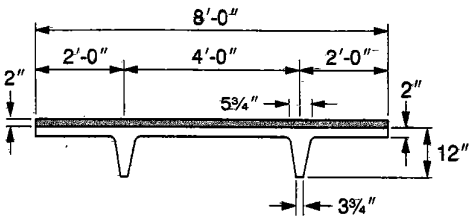
Safe loads shown include dead load of 10 psf for untopped members and 15 psf for topped members. Remainder is live load. Long-time cambers include superimposed dead load but do not include live load.

Key

186 — Safe superimposed service load, psf
0.2 — Estimated camber at erection, in.
0.3 — Estimated long-time camber, in.

DOUBLE TEE

8′-0″ x 12″
Lightweight Concrete



f'_c = 5,000 psi
f_{pu} = 270,000 psi

Section Properties

				Untopped		Topped	
A	=		in²	287		—	
I	=		in⁴	2,872		4,819	in⁴
y_b	=		in.	9.13		10.82	in.
y_t	=		in.	2.87		3.18	in.
S_b	=		in³	315		445	in³
S_t	=		in³	1,001		1,515	in³
wt	=		plf	229		429	plf
			psf	29		54	psf
V/S	=		in.	1.22			

8LDT12

Table of safe superimposed service load (psf) and cambers (in.)

No Topping

Strand Pattern	e_s, in. / e_c, in.	Span, ft															
		12	14	16	18	20	22	24	26	28	30	32	34	36	38	40	42
28-S	7.13	186	144	116	89	68	53	41	31								
	7.13	0.2	0.3	0.4	0.5	0.5	0.6	0.6	0.6								
		0.3	0.4	0.5	0.6	0.6	0.7	0.7	0.6								
48-S	5.13			195	153	120	95	77	62	50	41	33					
	5.13			0.6	0.7	0.8	1.0	1.1	1.2	1.3	1.3	1.3					
				0.8	0.9	1.1	1.2	1.3	1.3	1.4	1.3	1.2					
68-S	3.13				169	133	106	86	70	57	47	39	32				
	3.13				0.6	0.8	0.9	1.0	1.0	1.1	1.1	1.1	1.0				
					0.8	1.0	1.1	1.1	1.1	1.1	1.0	0.8	0.5				
68-D1	3.13											72	61	52	45	38	33
	6.63											2.6	2.8	3.0	3.1	3.2	3.2
												2.9	2.9	2.9	2.7	2.4	2.1

8LDT12+2

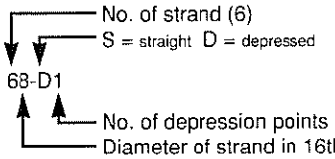
Table of safe superimposed service load (psf) and cambers (in.)

2″ Normal Weight Topping

Strand Pattern	e_s, in. / e_c, in.	Span, ft													
		12	14	16	18	20	22	24	26	28	30	32	34	36	38
28-S	7.13	207	157	123	91	66	47	33							
	7.13	0.2	0.3	0.4	0.5	0.5	0.6	0.6							
		0.2	0.3	0.4	0.4	0.4	0.3	0.2							
48-S	5.13				175	134	103	80	62	48	36				
	5.13				0.7	0.8	1.0	1.1	1.2	1.3	1.3				
					0.7	0.8	0.8	0.7	0.6	0.5	0.2				
68-S	3.13					162	126	99	79	62	47				
	3.13					0.8	0.9	1.0	1.0	1.1	1.1				
						0.7	0.7	0.6	0.5	0.3	0.1				
68-D1	3.13											72	59	46	33
	6.63											2.6	2.8	3.0	3.1
												1.3	1.0	0.6	0.0

Strength based on strain compatibility; bottom tension limited to $12\sqrt{f'_c}$; see pages 2-2–2-6 for explanation.
Shaded values require release strengths higher than 3500 psi.

Strand Pattern Designation



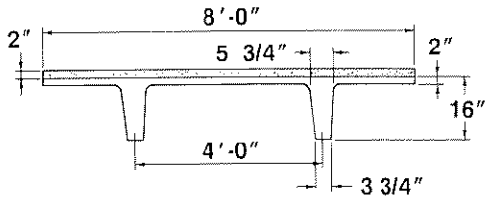
Safe loads shown include dead load of 10 psf for untopped members and 15 psf for topped members. Remainder is live load. Long-time cambers include superimposed dead load but do not include live load.

Key

168 — Safe superimposed service load, psf
0.2 — Estimated camber at erection, in.
0.3 — Estimated long-time camber, in.

DOUBLE TEE

8'-0" x 16"
Lightweight Concrete



f'_c = 5,000 psi

f_{pu} = 270,000 psi

Section Properties

		Untopped		Topped	
A	=	325	in.^{2}	—	
I	=	6,634	in.^{4}	10,094	in.^{4}
y_b	=	11.93	in.	13.99	in.
y_t	=	4.07	in.	4.01	in.
Z_b	=	556	in.^{3}	721	in.^{3}
Z_t	=	1,630	in.^{3}	2,517	in.^{3}
wt	=	260	plf	460	plf
		33	psf	58	psf
V/S	=	1.29	in.		

8LDT16

Table of safe superimposed service load (psf) and cambers

No Topping

Strand Pattern	e_e / e_c	Span, ft. 16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52	54	56
28-S	9.93	168	131	102	80	63	50	40	31													
	9.93	0.2	0.3	0.3	0.4	0.4	0.5	0.5	0.5													
		0.3	0.4	0.4	0.5	0.5	0.5	0.5	0.4													
48-S	8.93				167	137	113	94	78	66	56	47	40	33								
	8.93				0.8	0.9	1.0	1.2	1.3	1.4	1.4	1.5	1.5	1.5								
					1.0	1.2	1.3	1.4	1.5	1.6	1.6	1.6	1.5	1.4								
68-S	5.93				193	158	131	110	92	78	66	57	48	41	35	30						
	5.93				0.8	0.9	1.0	1.2	1.3	1.4	1.4	1.5	1.5	1.5	1.5	1.4						
					1.0	1.2	1.3	1.4	1.5	1.6	1.6	1.6	1.5	1.4	1.2	0.9						
68-D1	5.93						185	156	133	114	98	85	73	64	55	48	42	37	32			
	9.43						1.4	1.6	1.8	2.0	2.2	2.4	2.5	2.7	2.8	2.8	2.9	2.9	2.8			
							1.9	2.1	2.3	2.4	2.6	2.7	2.8	2.8	2.8	2.6	2.5	2.3	1.9			
88-D1	3.93																59	52	46	41	36	32
	9.18																3.7	3.8	3.9	3.9	3.9	3.8
																	3.7	3.5	3.3	3.0	2.5	2.0

8LDT16 + 2

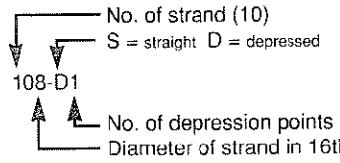
Table of safe superimposed service load (psf) and cambers

2" Normal Weight Topping

Strand Pattern	e_e / e_c	Span, ft. 16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48
28-S	9.93	175	133	99	74	55	41											
	9.93	0.2	0.3	0.3	0.4	0.4	0.5											
		0.3	0.3	0.3	0.3	0.3	0.2											
48-S	8.93				176	140	113	91	74	59	48	38						
	8.93				0.8	0.9	1.0	1.2	1.3	1.4	1.4	1.5						
					0.8	0.9	0.9	0.9	0.9	0.8	0.7	0.5						
68-S	5.93					172	140	115	94	77	63	52	42					
	5.93					0.9	1.0	1.2	1.3	1.4	1.4	1.5	1.5					
						0.9	0.9	0.9	0.9	0.8	0.7	0.5	0.2					
68-D1	5.93						195	162	135	114	96	80	68	57	47	39		
	9.43						1.4	1.6	1.8	2.0	2.2	2.4	2.5	2.7	2.8	2.8		
							1.4	1.5	1.6	1.6	1.6	1.5	1.4	1.1	0.8	0.3		
88-D1	3.93																52	41
	9.18																3.7	3.8
																	0.8	0.2

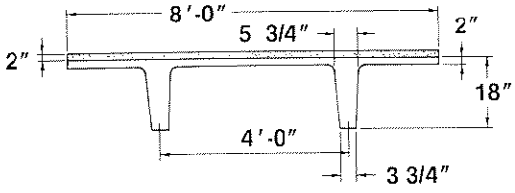
Strength based on strain compatibility; bottom tension limited to $12\sqrt{f'_c}$; see pages 2-3–2-6 for explanation.
Shaded values require release strengths higher than 3500 psi.

Strand Pattern Designation



DOUBLE TEE

8'-0" x 18"
Lightweight Concrete



Section Properties

		Untopped		Topped	
A	=	344	in.^{2}	—	
I	=	9,300	in.^{4}	13,799	in.^{4}
y_b	=	13.27	in.	15.51	in.
y_t	=	4.73	in.	4.49	in.
Z_b	=	701	in.^{3}	890	in.^{3}
Z_t	=	1,966	in.^{3}	3,073	in.^{3}
wt	=	275	plf	475	plf
		34	psf	59	psf
V/S	=	1.32	in.		

Safe loads shown include dead load of 10 psf for untopped members and 15 psf for topped members. Remainder is live load. Long-time cambers include superimposed dead load but do not include live load.

Key
194 — Safe superimposed service load, psf
0.2 — Estimated camber at erection, in.
0.3 — Estimated long-time camber, in.

f'_c = 5,000 psi
f_{pu} = 270,000 psi

8LDT18

Table of safe superimposed service load (psf) and cambers

No Topping

Strand Pattern	e_e / e_c	Span, ft. 16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52	54	56	58	60	62	64
28-S	11.27 11.27	194 0.2 0.3	152 0.2 0.3	118 0.3 0.4	93 0.3 0.4	74 0.4 0.4	59 0.4 0.5	47 0.4 0.4	38 0.4 0.4	30 0.4 0.4																
48-S	10.27 10.27				196 0.7 0.9	160 0.8 1.0	133 0.9 1.1	111 1.0 1.2	93 1.1 1.3	79 1.2 1.4	67 1.2 1.4	57 1.3 1.4	48 1.3 1.4	41 1.4 1.4	35 1.4 1.3										
68-S	7.27 7.27					195 0.8 1.1	162 0.9 1.2	136 1.1 1.3	115 1.2 1.4	98 1.3 1.5	84 1.4 1.6	72 1.4 1.6	62 1.5 1.6	53 1.5 1.6	46 1.5 1.5	39 1.5 1.4	34 1.5 1.1								
68-D1	7.27 10.77							184 1.4 1.8	156 1.6 2.0	134 1.7 2.2	116 1.9 2.3	101 2.1 2.5	88 2.2 2.6	76 2.4 2.6	67 2.5 2.6	59 2.6 2.6	51 2.6 2.5	45 2.6 2.4	39 2.6 2.2	34 2.6 2.0					
88-D1	5.27 10.52														92 3.1 3.6	81 3.2 3.6	72 3.4 3.7	64 3.5 3.6	57 3.6 3.5	51 3.7 3.4	45 3.8 3.1	40 3.7 2.8	36 3.7 2.4	31 3.5 1.9		
108-D1	3.27 10.27																						47 4.5 3.6	42 4.5 3.1	38 4.4 2.5	34 4.2 1.8

8LDT18 + 2

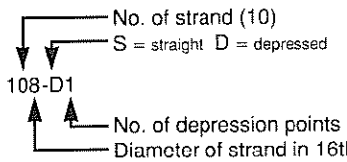
Table of safe superimposed service load (psf) and cambers

2" Normal Weight Topping

Strand Pattern	e_e / e_c	Span, ft. 16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	52
28-S	11.27 11.27	201 0.2 0.2	154 0.2 0.2	116 0.3 0.3	88 0.3 0.3	66 0.4 0.3	50 0.4 0.2	37 0.4 0.1												
48-S	10.27 10.27					164 0.8 0.8	133 0.9 0.8	108 1.0 0.8	88 1.1 0.9	72 1.2 0.8	59 1.2 0.7	47 1.3 0.6	38 1.3 0.4							
68-S	7.27 7.27						171 0.9 0.9	141 1.1 1.0	117 1.2 1.0	97 1.3 1.0	81 1.4 0.9	67 1.4 0.8	55 1.5 0.6	46 1.5 0.4	37 1.5 0.1					
68-D1	7.27 10.77							190 1.4 1.4	159 1.6 1.5	134 1.7 1.5	114 1.9 1.6	96 2.1 1.6	82 2.2 1.5	69 2.4 1.4	59 2.5 1.1	49 2.6 0.8	41 2.6 0.5	34 2.6 0.0		
88-D1	5.27 10.52														87 3.1 2.0	75 3.2 1.8	65 3.4 1.5	56 3.5 1.1	48 3.6 0.7	39 3.7 0.1

Strength based on strain compatibility; bottom tension limited to $12\sqrt{f'_c}$; see pages 2-3–2-6 for explanation.
Shaded values require release strengths higher than 3500 psi.

Strand Pattern Designation



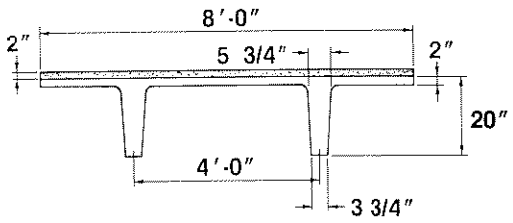
Safe loads shown include dead load of 10 psf for untopped members and 15 psf for topped members. Remainder is live load. Long-time cambers include superimposed dead load but do not include live load.

Key

184 — Safe superimposed service load, psf
0.7 — Estimated camber at erection, in.
0.8 — Estimated long-time camber, in.

DOUBLE TEE

8'-0" x 20"
Lightweight Concrete



f'_c = 5,000 psi

f_{pu} = 270,000 psi

Section Properties

		Untopped		Topped	
A	=	363	in.^{2}	—	
I	=	12,551	in.^{4}	18,278	in.^{4}
y_b	=	14.59	in.	17.02	in.
y_t	=	5.41	in.	4.98	in.
Z_b	=	860	in.^{3}	1,074	in.^{3}
Z_t	=	2,320	in.^{3}	3,670	in.^{3}
wt	=	290	plf	490	plf
		36	psf	61	psf
V/S	=	1.35	in.		

8LDT20

Table of safe superimposed service load (psf) and cambers

No Topping

Strand Pattern	e_e / e_c	Span, ft. 24	26	28	30	32	34	36	38	40	42	44	46	48	50	52	54	56	58	60	62	64	66	68
48-S	11.59	184	153	128	108	91	78	66	57	48	41	35												
	11.59	0.7	0.7	0.8	0.9	1.0	1.1	1.1	1.2	1.2	1.2	1.2												
		0.8	0.9	1.0	1.1	1.2	1.3	1.3	1.3	1.3	1.2	1.1												
68-S	8.59		193	162	138	118	101	87	75	65	57	49	42	37	32									
	8.59		0.9	1.0	1.1	1.2	1.2	1.3	1.4	1.5	1.5	1.5	1.5	1.5	1.4									
			1.1	1.2	1.3	1.4	1.5	1.6	1.6	1.6	1.6	1.5	1.4	1.2	1.0									
68-D1	8.59				180	155	134	117	102	89	78	69	61	53	47	41	36	32						
	12.09				1.4	1.5	1.6	1.8	2.0	2.1	2.2	2.3	2.3	2.4	2.4	2.4	2.4	2.3						
					1.8	1.9	2.0	2.2	2.3	2.4	2.4	2.4	2.4	2.3	2.2	2.1	1.9	1.6						
88-D1	6.59									137	121	107	95	85	76	68	60	54	48	43	38	34	30	
	11.84									2.4	2.6	2.8	2.9	3.1	3.2	3.4	3.5	3.5	3.5	3.5	3.5	3.4	3.2	
										3.0	3.2	3.3	3.4	3.5	3.5	3.5	3.5	3.3	3.1	2.8	2.5	2.1	1.7	
108-D1	4.59																	63	57	51	46	42	37	34
	11.59																	1.8	1.7	4.4	4.4	4.4	4.3	4.2
																		4.2	4.1	3.8	3.5	3.0	2.5	1.9

8LDT20 + 2

Table of safe superimposed service load (psf) and cambers

2" Normal Weight Topping

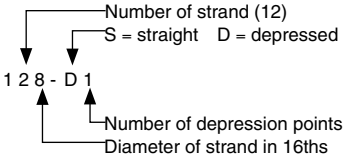
Strand Pattern	e_e / e_c	Span, ft. 24	26	28	30	32	34	36	38	40	42	44	46	48	50	52	54	56	58
48-S	11.59	188	153	125	103	85	70	57	46	31									
	11.59	0.7	0.7	0.8	0.9	1.0	1.1	1.1	1.2	1.2									
		0.7	0.7	0.8	0.8	0.8	0.7	0.7	0.6	0.4									
68-S	8.59			167	140	117	98	82	69	58	48	39							
	8.59			1.0	1.1	1.2	1.2	1.3	1.4	1.5	1.5	1.5							
				0.9	1.0	1.0	1.0	0.9	0.8	0.7	0.5	0.2							
68-D1	8.59				182	154	131	112	96	82	70	59	50	42	35				
	12.09				1.4	1.5	1.6	1.8	2.0	2.1	2.2	2.3	2.3	2.4	2.4				
					1.4	1.4	1.4	1.5	1.5	1.4	1.3	1.1	0.8	0.5	0.1				
88-D1	6.59								136	118	102	89	77	67	58	50	43	36	
	11.84								2.4	2.6	2.8	2.9	3.1	3.2	3.4	3.5	3.5	3.5	
									2.1	2.1	2.0	2.0	1.8	1.6	1.3	0.9	0.4	0.2	
108-D1	4.59																	53	45
	11.59																	4.3	4.3
																		0.8	0.2

Strength based on strain compatibility; bottom tension limited to $12\sqrt{f'_c}$; see pages 2-3–2-6 for explanation.
Shaded values require release strengths higher than 3500 psi.

3.4 Double-Tee Load Tables

3

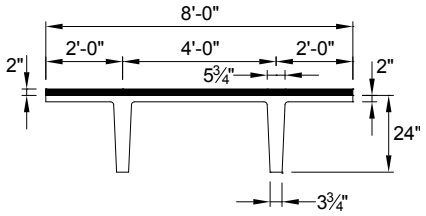
Strand Pattern Designation



Safe loads shown include dead load of 10 lb/ft² for untopped members and 15 lb/ft² for topped members. Remainder is live load. Long-time cambers include superimposed dead load but do not include live load.

Key
186 – Safe superimposed service load, lb/ft²
0.7 – Estimated camber at erection, in.
0.9 – Estimated long-time camber, in.

8'-0" × 24"
Normalweight Concrete



f'_c = 5000 psi

f_{pu} = 270,000 psi

$^1/_2$-in.-diameter regular strand

Check with regional producers for availability.

Section Properties

		No Topping		2 in. Topping	
A	=	401	in.²	—	
I	=	20,985	in.⁴	27,720	in.⁴
y_b	=	17.15	in.	19.27	in.
y_t	=	6.85	in.	6.73	in.
S_b	=	1224	in.³	1439	in.³
S_t	=	3064	in.³	4119	in.³
wt	=	418	lb/ft	618	lb/ft
DL	=	52	lb/ft²	77	lb/ft²
V/S	=	1.41	in.		

8DT24

Table of safe superimposed service load, lb/ft², and cambers, in. **No Topping**

Strand pattern	y_s(end) y_s(center) in.	Span, ft																						
		32	34	36	38	40	42	44	46	48	50	52	54	56	58	60	62	64	66	68	70	72	74	76
68-S	4.00	186	161	140	122	106	93	81	71	62	55	48	42	36	31	27								
	4.00	0.7	0.8	0.8	0.9	0.9	1.0	1.0	1.0	1.0	1.0	0.9	0.8	0.7	0.6	0.5								
		0.9	1.0	1.1	1.1	1.1	1.1	1.1	1.1	1.0	0.9	0.8	0.6	0.4	0.1	–0.2								
88-S	5.00			185	162	143	126	112	99	88	78	70	62	55	49	43	38	33	29					
	5.00			1.1	1.2	1.3	1.3	1.4	1.4	1.5	1.5	1.5	1.5	1.4	1.4	1.3	1.1	1.0	0.8					
				1.4	1.5	1.6	1.6	1.7	1.7	1.7	1.7	1.6	1.5	1.4	1.2	0.9	0.6	0.3	–0.2					
108-S	6.00				197	174	155	138	123	110	98	88	79	71	64	57	51	45	39	34	29			
	6.00				1.4	1.5	1.6	1.7	1.7	1.8	1.9	1.9	1.9	1.9	1.9	1.9	1.8	1.7	1.5	1.3	1.1			
					1.8	1.9	2.0	2.1	2.1	2.1	2.1	2.1	2.1	2.0	1.9	1.8	1.6	1.3	0.9	0.5	0.0			
128-S	7.00							159	142	128	114	101	90	81	73	66	59	53	47	42	37	32	27	
	7.00							1.8	1.9	2.0	2.1	2.1	2.2	2.2	2.2	2.2	2.2	2.1	2.0	1.9	1.7	1.5	1.2	
								2.3	2.4	2.4	2.5	2.5	2.4	2.4	2.3	2.2	2.1	1.8	1.6	1.2	0.8	0.3	–0.3	
128-D1	11.67												114	103	92	83	74	66	59	53	48	43	39	35
	3.25												2.5	2.6	2.6	2.7	2.7	2.6	2.6	2.5	2.3	2.1	1.9	1.6
													2.9	2.9	2.9	2.8	2.6	2.4	2.2	1.8	1.5	1.1	0.6	0.0
148-D1	12.86																		71	64	58	52	46	42
	3.50																		3.1	3.1	3.0	2.9	2.7	2.5
																			3.0	2.7	2.4	2.0	1.6	1.1

8DT24 + 2

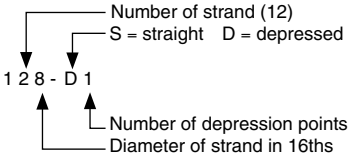
Table of safe superimposed service load, lb/ft², and cambers, in. **2 in. Normalweight Topping**

Strand pattern	y_s(end) y_s(center) in.	Span, ft																					
		28	30	32	34	36	38	40	42	44	46	48	50	52	54	56	58	60	62	64	66	68	70
48-S	3.00	169	141	117	97	81	67	55	45	36	29												
	3.00	0.4	0.4	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.4												
		0.4	0.4	0.4	0.4	0.3	0.3	0.2	0.0	–0.1	–0.3												
68-S	4.00			189	161	138	118	101	87	74	64	54	45	38	30								
	4.00			0.7	0.8	0.8	0.9	0.9	1.0	1.0	1.0	1.0	1.0	0.9	0.8								
				0.7	0.8	0.8	0.7	0.7	0.6	0.6	0.4	0.2	0.0	–0.3	–0.6								
88-S	5.00					188	163	142	124	108	94	82	71	62	52	42	33						
	5.00					1.1	1.1	1.3	1.3	1.4	1.4	1.5	1.5	1.5	1.5	1.4	1.4						
						1.1	1.2	1.1	1.1	1.1	1.0	0.9	0.7	0.5	0.3	–0.1	–0.5						
108-S	6.00							176	155	136	120	104	90	77	66	56	47	39	31				
	6.00							1.5	1.6	1.7	1.7	1.8	1.9	1.9	1.9	1.9	1.9	1.9	1.8				
								1.4	1.4	1.4	1.4	1.3	1.2	1.0	0.8	0.6	0.3	–0.1	–0.6				
128-S	7.00									160	140	121	104	90	77	67	57	48	41	33	27		
	7.00									1.8	1.9	2.0	2.1	2.1	2.2	2.2	2.2	2.2	2.2	2.1	2.0		
										1.6	1.6	1.6	1.5	1.3	1.1	0.9	0.6	0.3	–0.1	–0.6	–1.1		
128-D1	11.67														105	91	79	68	58	49	40	34	28
	3.25														2.5	2.6	2.6	2.7	2.7	2.6	2.6	2.5	2.3
															1.6	1.4	1.1	0.8	0.4	0.0	–0.6	–1.2	–1.8

Strength is based on strain compatibility; bottom tension is limited to $12\sqrt{f'_c}$; see pages 3-8 through 3-11 for explanation.
Shaded values require release strengths higher than 3500 psi.

3.4 Double-Tee Load Tables (cont.)

Strand Pattern Designation

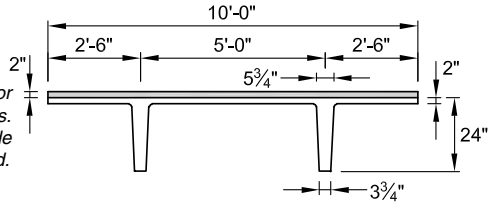


Safe loads shown include dead load of 10 lb/ft² for untopped members and 15 lb/ft² for topped members. Remainder is live load. Long-time cambers include superimposed dead load but do not include live load.

Key
171 – Safe superimposed service load, lb/ft²
0.6 – Estimated camber at erection, in.
0.8 – Estimated long-time camber, in.

Check with regional producers for availability.

10'-0" × 24"
Normalweight Concrete



f'_c = 5000 psi

f_{pu} = 270,000 psi

1/2-in.-diameter regular strand

Section Properties

		No Topping		2 in. Topping	
A	=	449	in.²	–	
I	=	22,469	in.⁴	29,395	in.⁴
y_b	=	17.77	in.	19.89	in.
y_t	=	6.23	in.	6.11	in.
S_b	=	1264	in.³	1478	in.³
S_t	=	3607	in.³	4812	in.³
wt	=	468	lb/ft	718	lb/ft
DL	=	47	lb/ft²	72	lb/ft²
V/S	=	1.35	in.		

10DT24

Table of safe superimposed service load, lb/ft², and cambers, in. **No Topping**

Strand pattern	y_s(end) y_s(center) in.	Span, ft																								
		30	32	34	36	38	40	42	44	46	48	50	52	54	56	58	60	62	64	66	68	70	72	74	76	78
68-S	4.00	171	146	126	109	94	82	71	62	54	47	41	35	30	26											
	4.00	0.6	0.7	0.7	0.8	0.8	0.9	0.9	0.9	0.9	0.9	0.8	0.8	0.7	0.6											
		0.8	0.9	0.9	1.0	1.0	1.0	1.0	1.0	0.9	0.8	0.7	0.5	0.3	0.0											
88-S	5.00		193	167	146	127	112	98	87	77	68	60	53	47	41	36	32	27								
	5.00		0.9	1.0	1.1	1.1	1.2	1.3	1.3	1.4	1.4	1.4	1.4	1.3	1.3	1.2	1.0	0.9								
			1.2	1.3	1.4	1.4	1.5	1.5	1.5	1.5	1.5	1.5	1.4	1.2	1.0	0.8	0.5	0.1								
108-S	6.00				177	156	137	121	108	96	85	76	68	61	54	48	43	38	33	29						
	6.00				1.2	1.3	1.4	1.5	1.6	1.7	1.7	1.8	1.8	1.8	1.8	1.8	1.7	1.6	1.4	1.2						
					1.6	1.7	1.8	1.9	1.9	2.0	2.0	2.0	1.9	1.9	1.8	1.6	1.4	1.1	0.7	0.3						
128-S	7.00						159	141	125	112	100	90	80	72	64	58	52	46	41	36	31	26				
	7.00						1.6	1.7	1.8	1.9	1.9	2.0	2.0	2.1	2.1	2.1	2.1	2.0	1.9	1.8	1.6	1.4				
							2.0	2.1	2.2	2.3	2.3	2.3	2.3	2.2	2.2	2.1	1.9	1.7	1.4	1.1	0.6	0.1				
128-D1	11.67												100	90	80	72	64	57	51	46	41	37	33	30	26	
	3.25												2.3	2.4	2.5	2.5	2.5	2.5	2.4	2.3	2.2	2.0	1.6	1.5	1.2	
													2.7	2.7	2.6	2.5	2.4	2.2	1.9	1.6	1.3	0.9	0.4	-0.2	-0.9	
148-D1	12.86																	68	61	55	49	43	39	36	32	29
	3.50																	2.9	2.9	2.9	2.8	2.7	2.6	2.4	2.1	1.8
																		2.9	2.7	2.5	2.2	1.8	1.4	0.9	0.3	-0.3

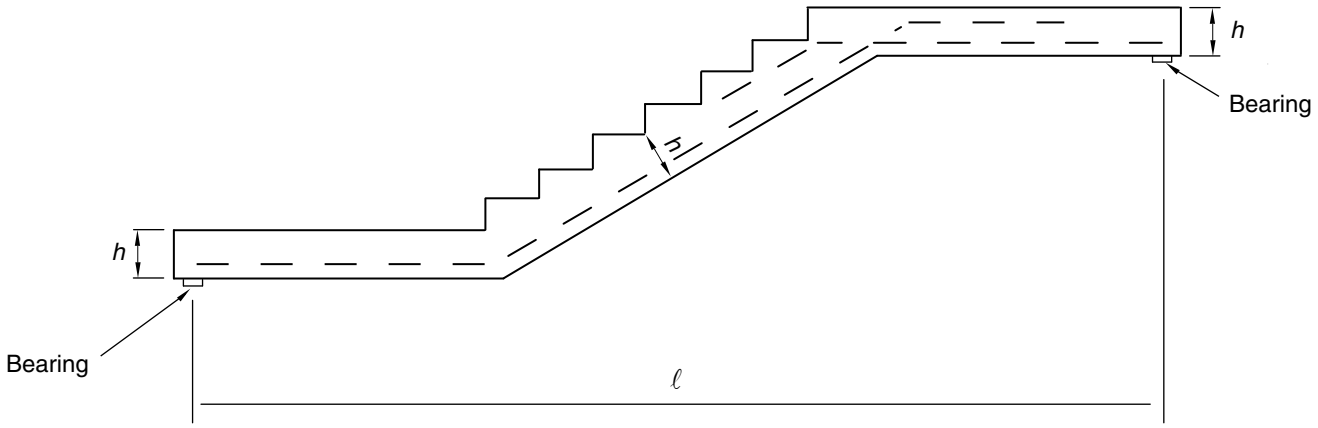
10DT24 + 2

Table of safe superimposed service load, lb/ft², and cambers, in. **2 in. Normalweight Topping**

Strand pattern	y_s(end) y_s(center) in.	Span, ft																		
		30	32	34	36	38	40	42	44	46	48	50	52	54	56	58	60	62	64	66
68-S	4.00	172	145	123	104	89	75	63	56	45	37	30								
	4.00	0.6	0.7	0.7	0.8	0.8	0.9	0.9	0.9	0.9	0.9	0.8								
		0.6	0.7	0.7	0.7	0.6	0.6	0.5	0.4	0.2	0.0	–0.3								
88-S	5.00		197	169	145	125	108	93	83	69	60	51	43	34						
	5.00		0.9	1.0	1.1	1.1	1.2	1.3	1.3	1.4	1.4	1.4	1.4	1.3						
			0.9	1.0	1.0	1.0	1.0	0.9	0.9	0.7	0.6	0.4	0.2	–0.2						
108-S	6.00				180	156	136	119	104	91	79	68	57	48	40	31				
	6.00				1.2	1.3	1.4	1.5	1.6	1.7	1.7	1.8	1.8	1.8	1.8	1.8				
					1.2	1.3	1.3	1.3	1.2	1.1	1.0	0.9	0.7	0.5	0.2	–0.2				
128-S	7.00						160	141	124	109	93	79	68	58	48	40	33			
	7.00						1.6	1.7	1.8	1.9	1.9	2.0	2.1	2.1	2.1	2.1	2.1			
							1.5	1.5	1.5	1.4	1.3	1.2	1.0	0.8	0.5	0.2	–0.2			
128-D1	11.67												92	79	67	57	47	39	32	
	3.25												2.3	2.4	2.5	2.5	2.5	2.5	2.4	
													1.4	1.2	0.9	0.6	0.3	–0.2	–1.7	
148-D1	12.86																	53	44	36
	3.50																	2.9	2.9	2.9
																		0.5	0.0	–0.6

Strength is based on strain compatibility; bottom tension is limited to $12\sqrt{f'_c}$; see pages 3–8 through 3–11 for explanation.
Shaded values require release strengths higher than 3500 psi.

Design Aid 3.12.10 Recommended Maximum Spans for Precast Concrete Stairs



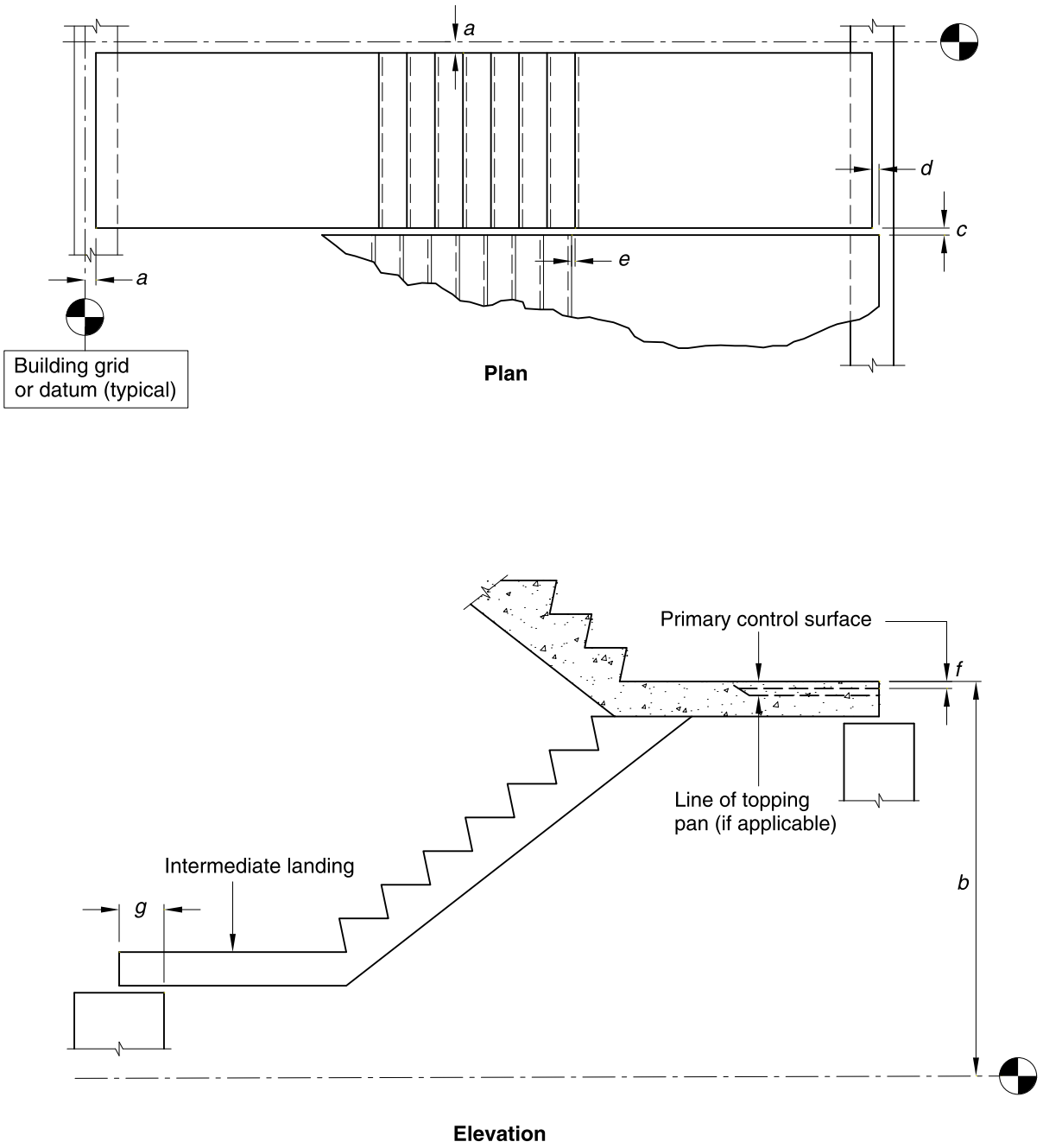
— — — Typical main reinforcing pattern

Thickness *h*, in.	Span ℓ, ft
6	14
8	19
10	23
12	26

Notes:

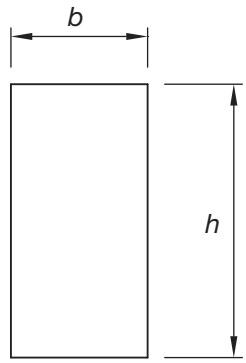
1. Stair reinforcing must be designed for all factored loads.
2. Normalweight concrete.
3. Recommended maximum span ℓ is based on dead weight of stair plus 100 lb/ft^2 live load.
4. Minimum concrete 28-day compressive strength is 5000 psi.
5. When deflections are required to be checked in accordance with ACI 318-05 Section 9.5, cracked section properties are required to be used. Top steel may be required to control deflections.
6. Longer spans can be obtained with higher concrete strength or special reinforcing.

Fig. 13.3.9 Stair unit erection tolerance



3.9 Rectangular Beam Load Tables

Normalweight concrete



f'_c = 5000 psi
f_{pu} = 270,000 psi
½ in. diameter,
low-relaxation strand

Section Properties							
Designation	***b*** **in.**	***h*** **in.**	***A*** **in.²**	***I*** **in.⁴**	y_b **in.**	***S*** **in.³**	***wt*** **lb/ft**
12RB16	12	16	192	4096	8	512	200
12RB20	12	20	240	8000	10	800	250
12RB24	12	24	288	13,824	12	1152	300
12RB28	12	28	336	21,952	14	1568	350
12RB32	12	32	384	32,768	16	2048	400
12RB36	12	36	432	46,656	18	2592	450
16RB24	16	24	384	18,432	12	1536	400
16RB28	16	28	448	29,269	14	2091	467
16RB32	16	32	512	43,691	16	2731	533
16RB36	16	36	576	62,208	18	3456	600
16RB40	16	40	640	85,333	20	4267	667

1. Check local area for availability of other sizes.
2. Loads shown include 50% superimposed dead load and 50% live load. Top tension stress at transfer has been allowed to exceed $6\sqrt{f'_c}$; therefore, top reinforcement is required.
3. Loads can be significantly increased by use of structural composite topping.

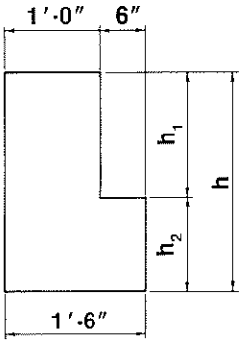
Key
3550 – Superimposed service load capacity, lb/ft
0.4 – Estimated camber at erection, in.
0.2 – Estimated long-time camber, in.

Table of superimposed service load capacity, lb/ft, and cambers, in.

Designation	Number strand	y_s in.	Span, ft																		
			16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50	
12RB16	5	3.00	3550 0.4 0.2	2770 0.5 0.2	2210 0.6 0.2	1790 0.8 0.2	1480 0.9 0.3	1230 1.0 0.3	1040 1.1 0.3												
12RB20	8	3.00	6160 0.4 0.2	4820 0.5 0.2	3860 0.6 0.3	3150 0.7 0.3	2620 0.9 0.4	2200 1.0 0.4	1860 1.1 0.4	1600 1.3 0.5	1380 1.4 0.5	1190 1.5 0.5	1040 1.7 0.5								
12RB24	10	3.60	8950 0.4 0.2	7010 0.4 0.2	5630 0.5 0.3	4610 0.7 0.3	3830 0.8 0.3	3230 0.9 0.4	2740 1.0 0.4	2360 1.1 0.5	2040 1.3 0.5	1780 1.4 0.6	1560 1.5 0.6	1370 1.6 0.6	1210 1.8 0.6	1070 1.9 0.7	960 2.0 0.6				
12RB28	12	4.00		9780 0.4 0.2	7860 0.5 0.2	6440 0.6 0.3	5370 0.7 0.3	4530 0.8 0.4	3860 0.9 0.4	3320 1.0 0.5	2890 1.2 0.5	2520 1. 3 0.6	2220 1.4 0.6	1960 1.5 0.7	1740 1.7 0.7	1550 1.8 0.7	1380 1.9 0.8	1240 2.0 0.8	1110 2.1 0.8	1000 2.2 0.8
12RB32	13	4.77				8320 0.5 0.2	6930 0.6 0.3	5850 0.7 0.3	5000 0.8 0.3	4310 0.9 0.4	3750 1.0. 0.4	3280 1.1 0.4	2890 1.2 0.5	2560 1.3 0.5	2270 1.4 0.5	2030 1.5 0.5	1820 1.6 0.6	1630 1.7 0.6	1470 1.8 0.6	1330 1.9 0.6
12RB36	15	5.07					9010 0.5 0.2	7620 0.6 0.3	6520 0.7 0.3	5630 0.8 0.4	4900 0.9 0.4	4290 1.0 0.4	3790 1.1 0.5	3360 1.2 0.5	2990 1.3 0.6	2680 1.4 0.6	2410 1.5 0.6	2170 1.6 0.6	1960 1.7 0.7	1780 1.8 0.7
16RB24	13	3.54		9390 0.4 0.2	7540 0.5 0.2	6170 0.6 0.3	5130 0.8 0.3	4320 0.9 0.4	3680 1.0 0.4	3160 1.1 0.5	2730 1.2 0.5	2380 1.4 0.5	2090 1.5 0.5	1840 1.6 0.6	1620 1.7 0.6	1440 1.8 0.6	1280 1.9 0.6	1140 2.0 0.6	1020 2.1 0.5	
16RB28	14	3.71				8730 0.5 0.2	7270 0.6 0.2	6130 0.7 0.3	5230 0.8 0.3	4510 0.9 0.3	3910 1.1 0.4	3420 1.2 0.4	3010 1.3 0.4	2660 1.4 0.4	2360 1.5 0.4	2100 1.6 0.4	1880 1.7 0.4	1680 1.8 0.4	1510 1.9 0.4	1360 1.9 0.3
16RB32	18	4.67					9340 0.6 0.3	7890 0.7 0.3	6740 0.8 0.4	5810 0.9 0.4	5050 1.0 0.5	4420 1.1 0.5	3890 1.2 0.5	3450 1.3 0.6	3070 1.5 0.6	2740 1.6 0.6	2450 1.7 0.7	2210 1.8 0.7	1990 1.9 0.7	1800 2.0 0.7
16RB36	20	5.40						9940 0.6 0.3	5800 0.7 0.3	7340 0.8 0.4	6390 0.9 0.4	5600 1.0 0.4	4940 1.1 0.5	4380 1.2 0.5	3900 1.3 0.5	3490 1.4 0.6	3130 1.5 0.6	2820 1.6 0.6	2550 1.7 0.6	2310 1.8 0.6
16RB40	22	6.00								9120 0.7 0.3	7940 0.8 0.4	6970 0.9 0.4	6160 1.0 0.4	5470 1.1 0.5	4880 1.2 0.5	4370 1.3 0.5	3930 1.4 0.5	3550 1.5 0.6	3210 1.6 0.6	2910 1.7 0.6

Strength is based on strain compatibility; bottom tension is limited to $12\sqrt{f'_c}$; see pages 3–8 through 3–11 for explanation.

L-SHAPED BEAMS



f'_c = 5,000 psi

f_{pu} = 270,000 psi

½ in. diameter
low-relaxation strand

Key

6,675 — Safe superimposed service load, plf
0.3 — Estimated camber at erection, in.
0.1 — Estimated long-time camber, in.

Normal Weight Concrete

Section Properties								
Designation	h (in.)	h_1/h_2 (in.)	A (in.²)	I (in.⁴)	y_b (in.)	Z_b (in.³)	Z_t (in.³)	wt (plf)
18LB20	20	12/8	288	9,696	9.00	1,077	882	300
18LB24	24	12/12	360	16,762	10.80	1,552	1,270	375
18LB28	28	16/12	408	26,611	12.59	2,114	1,727	425
18LB32	32	20/12	456	39,695	14.42	2,753	2,258	475
18LB36	36	24/12	504	56,407	16.29	3,463	2,862	525
18LB40	40	24/16	576	77,568	18.00	4,309	3,526	600
18LB44	44	28/16	624	103,153	19.85	5,197	4,271	650
18LB48	48	32/16	672	133,705	21.71	6,159	5,086	700
18LB50	52	36/16	720	169,613	23.60	7,187	5,972	750
18LB56	56	40/16	768	211,264	25.50	8,285	6,927	800
18LB60	60	44/16	816	259,046	27.41	9,451	7,949	850

1. Check local area for availability of other sizes.

2. Safe loads shown include 50% dead load and 50% live load. 800 psi top tension has been allowed, therefore additional top reinforcement is required.

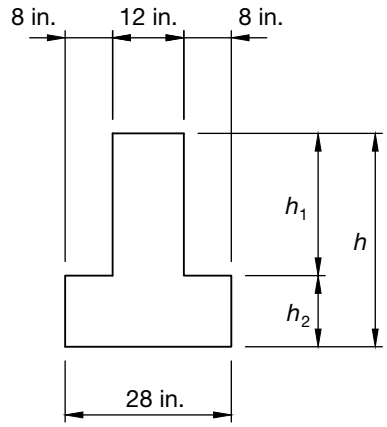
3. Safe loads can be significantly increased by use of structural composite topping.

Table of safe superimposed service load (plf) and cambers

Designation	No. Strand	e	Span, ft.																	
			16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50
18LB20	9	6.26	6,675 0.3 0.1	5,211 0.4 0.1	4,164 0.5 0.2	3,389 0.6 0.2	2,800 0.7 0.2	2,341 0.8 0.2	1,978 0.9 0.2	1,684 1.0 0.2	1,444 1.1 0.2	1,245 1.2 0.2	1,080 1.2 0.2							
18LB24	10	7.67	9,625 0.3 0.1	7,534 0.3 0.1	6,038 0.4 0.1	4,931 0.5 0.1	4,090 0.6 0.2	3,434 0.7 0.2	2,915 0.7 0.2	2,495 0.8 0.2	2,152 0.9 0.2	1,868 1.0 0.2	1,629 1.1 0.2	1,427 1.1 0.1	1,253 1.2 0.1	1,104 1.2 0.1	975 1.3 0.0			
18LB28	12	8.93			8,387 0.4 0.1	6,857 0.4 0.1	5,694 0.5 0.2	4,789 0.6 0.2	4,071 0.7 0.2	3,491 0.8 0.2	3,017 0.8 0.2	2,624 0.9 0.2	2,295 1.0 0.2	2,017 1.1 0.2	1,781 1.1 0.2	1,578 1.2 0.2	1,402 1.3 0.2	1,249 1.3 0.1	1,114 1.4 0.1	995 1.4 0.1
18LB32	14	10.22				9,049 0.4 0.1	7,528 0.5 0.2	6,344 0.5 0.2	5,404 0.6 0.2	4,647 0.7 0.2	4,026 0.8 0.2	3,512 0.8 0.2	3,082 0.9 0.2	2,717 1.0 0.3	2,406 1.1 0.3	2,138 1.1 0.3	1,906 1.2 0.2	1,706 1.3 0.2	1,530 1.3 0.2	1,375 1.4 0.2
18LB36	16	11.52					9,617 0.4 0.1	8,117 0.5 0.2	6,927 0.6 0.2	5,966 0.6 0.2	5,180 0.7 0.2	4,529 0.8 0.2	3,983 0.9 0.3	3,521 0.9 0.3	3,126 1.0 0.3	2,787 1.1 0.3	2,493 1.2 0.3	2,236 1.2 0.3	2,011 1.3 0.3	1,813 1.3 0.3
18LB40	18	12.52							8,581 0.5 0.2	7,398 0.6 0.2	6,429 0.6 0.2	5,626 0.7 0.2	4,954 0.8 0.2	4,385 0.8 0.2	3,899 0.9 0.3	3,480 1.0 0.3	3,118 1.1 0.3	2,802 1.1 0.3	2,524 1.2 0.3	2,281 1.2 0.3
18LB44	19	14.19								9,039 0.5 0.2	7,866 0.6 0.2	6,893 0.6 0.2	6,078 0.7 0.2	5,389 0.8 0.2	4,800 0.8 0.2	4,293 0.9 0.2	3,854 1.0 0.2	3,471 1.0 0.3	3,135 1.1 0.3	2,838 1.2 0.2
18LB48	21	15.48									9,439 0.5 0.2	8,281 0.6 0.2	7,311 0.7 0.2	6,490 0.7 0.2	5,789 0.8 0.2	5,186 0.9 0.3	4,663 0.9 0.3	4,207 1.0 0.3	3,806 1.1 0.3	3,453 1.1 0.3
18LB52	23	16.78										9,798 0.6 0.2	8,658 0.6 0.2	7,694 0.7 0.2	6,871 0.8 0.2	6,162 0.8 0.3	5,548 0.9 0.3	5,012 1.0 0.3	4,542 1.0 0.3	4,127 1.1 0.3
18LB56	25	18.07												8,993 0.7 0.2	8,038 0.7 0.2	7,216 0.8 0.3	6,504 0.8 0.3	5,883 0.9 0.3	5,337 1.0 0.3	4,856 1.0 0.3
18LB60	27	19.36													9,292 0.7 0.2	8,349 0.7 0.3	7,532 0.8 0.3	6,819 0.9 0.3	6,193 0.9 0.3	5,641 1.0 0.3

3.11 Inverted T Beam Load Tables

3



f'_c = 5000 psi
f_{pu} = 270,000 psi
½ in. diameter,
low-relaxation strand

Normalweight concrete

Section Properties								
Designation	***h*** **in.**	***h_1/h_2*** **in.**	***A*** **in.²**	***I*** **in.⁴**	***y_b*** **in.**	***S_b*** **in.³**	***S_t*** **in.³**	***wt*** **lb/ft**
28IT20	20	12/8	368	11,688	7.91	1478	967	383
28IT24	24	12/12	480	20,275	9.60	2112	1408	500
28IT28	28	16/12	528	32,076	11.09	2892	1897	550
28IT32	32	20/12	576	47,872	12.67	3778	2477	600
28IT36	36	24/12	624	68,101	14.31	4759	3140	650
28IT40	40	24/16	736	93,503	15.83	5907	3869	767
28IT44	44	28/16	784	124,437	17.43	7139	4683	817
28IT48	48	32/16	832	161,424	19.08	8460	5582	867
28IT52	52	36/16	880	204,884	20.76	9869	6558	917
28IT56	56	40/16	928	255,229	22.48	11,354	7614	967
28IT60	60	44/16	976	312,866	24.23	12,912	8747	1017

1. Check local area for availability of other sizes.
2. Loads shown include 50% superimposed dead load and 50% live load. Top tension stress at transfer has been allowed to exceed 6 $\sqrt{f'_c}$; therefore, top reinforcement is required.
3. Loads can be significantly increased by use of structural composite topping.

Key
6510 – Superimposed service load capacity, lb/ft
0.2 – Estimated camber at erection, in.
0.1 – Estimated long-time camber, in.

Table of superimposed service load capacity, lb/ft, and cambers, in.

Designation	Number strand	y_s in.	Span, ft																	
			16	18	20	22	24	26	28	30	32	34	36	38	40	42	44	46	48	50
28IT20	9	2.44	6510 0.2 0.1	5070 0.3 0.1	4040 0.4 0.1	3280 0.4 0.1	2710 0.5 0.1	2260 0.5 0.1	1900 0.6 0.0	1610 0.7 0.0	1380 0.7 0.0	1180 0.7 0.0	1020 0.8 –0.1							
28IT24	18	2.73	9610 0.2 0.1	7500 0.3 0.1	5990 0.3 0.1	4880 0.4 0.1	4030 0.4 0.1	3370 0.5 0.1	2850 0.6 0.1	2420 0.6 0.1	2080 0.7 0.1	1790 0.7 0.1	1550 0.7 0.0	1350 0.8 0.0	1170 0.8 –0.1	1020 0.8 –0.2				
28IT28	13	3.08			8350 0.3 0.1	6820 0.3 0.1	5650 0.4 0.1	4750 0.5 0.1	4030 0.5 0.1	3450 0.6 0.1	2970 0.6 0.1	2580 0.7 0.1	2250 0.7 0.1	1970 0.8 0.1	1730 0.8 0.0	1530 0.8 0.0	1350 0.9 –0.1	1190 0.8 –0.2	1060 0.8 –0.2	
28IT32	15	3.47				9040 0.3 0.1	7520 0.4 0.1	5330 0.4 0.1	5380 0.5 0.1	4620 0.5 0.1	4000 0.6 0.1	3490 0.6 0.1	3050 0.7 0.1	2690 0.7 0.1	2370 0.8 0.1	2110 0.8 0.1	1870 0.9 0.0	1670 0.9 0.0	1490 0.9 0.0	1330 0.9 –0.1
28IT36	16	3.50					9830 0.3 0.1	8290 0.4 0.1	7070 0.4 0.1	6090 0.5 0.1	5280 0.5 0.1	4610 0.6 0.1	4060 0.6 0.1	3580 0.7 0.1	3180 0.7 0.1	2830 0.8 0.1	2530 0.8 0.0	2270 0.9 0.0	2040 0.9 0.0	1830 0.9 –0.1
28IT40	19	4.21							8630 0.4 0.1	7440 0.5 0.1	6460 0.5 0.1	5640 0.6 0.1	4960 0.6 0.1	4390 0.7 0.1	3890 0.7 0.1	3470 0.8 0.1	3100 0.8 0.1	2780 0.8 0.1	2500 0.9 0.1	2250 0.9 0.1
28IT44	20	4.40								9180 0.4 0.1	7980 0.5 0.1	6990 0.5 0.1	6160 0.6 0.1	5460 0.6 0.1	4860 0.7 0.1	4340 0.7 0.1	3890 0.7 0.1	3500 0.8 0.1	3160 0.8 0.1	2850 0.8 0.0
28IT48	22	4.55									9710 0.4 0.1	8520 0.5 0.1	7520 0.5 0.1	6670 0.6 0.1	5950 0.6 0.1	5330 0.7 0.1	4790 0.7 0.1	4320 0.8 0.1	3900 0.8 0.1	3540 0.9 0.1
28IT52	24	5.17										9980 0.5 0.1	8820 0.5 0.1	7830 0.6 0.1	6990 0.6 0.1	6270 0.6 0.1	5640 0.7 0.1	4100 0.7 0.1	4610 0.8 0.1	4190 0.8 0.1
28IT56	26	5.23												9300 0.5 0.2	8310 0.6 0.2	7460 0.6 0.2	6730 0.7 0.2	6080 0.7 0.2	5520 0.8 0.2	5020 0.8 0.2
28IT60	28	5.57													9640 0.6 0.2	8660 0.6 0.2	7820 0.7 0.2	7080 0.7 0.2	6430 0.8 0.2	5850 0.8 0.2

Strength is based on strain compatibility; bottom tension is limited to 12 $\sqrt{f'_c}$; see pages 3–8 through 3–11 for explanation.

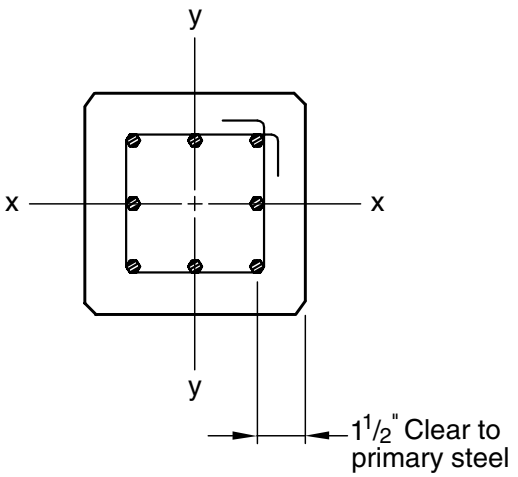
Design Aid 3.12.2. Design Strength Interaction Curves for Precast, Reinforced Concrete Columns

CRITERIA

1. Concrete f'_c = 5000 psi, normalweight
2. Reinforcement f_y = 60,000 psi
3. Curves shown for full development of reinforcement
4. Horizontal portion of curve is the maximum for tied columns = $0.80\phi P_O$
5. Varies linearly from 0.9 for tension-controlled sections to 0.65 for compression-controlled sections in accordance with ACI 318-05 Section 9.3.2

USE OF CURVES

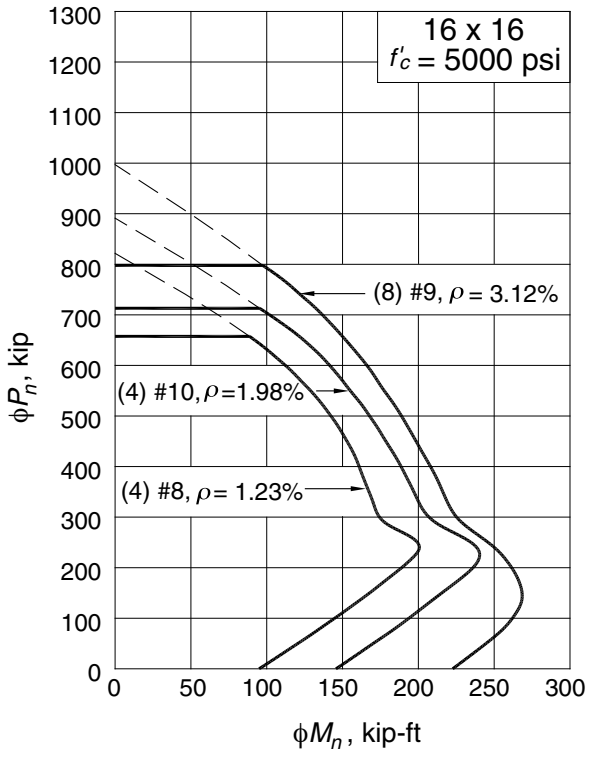
1. Enter at left with applied factored axial load P_u
2. Enter at bottom with applied magnified factored moment δM_u
3. Intersection point must be to the left of curve indicating required reinforcement.

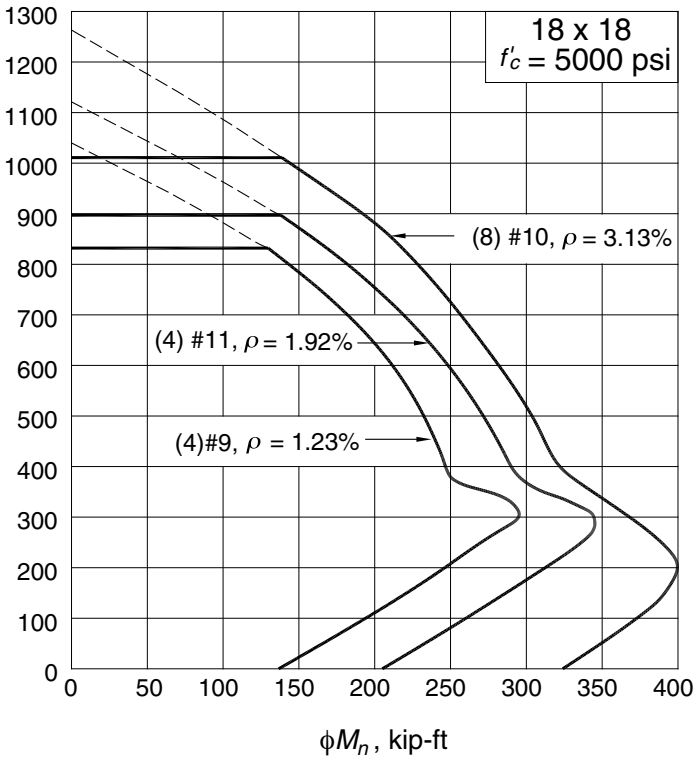


NOTATION

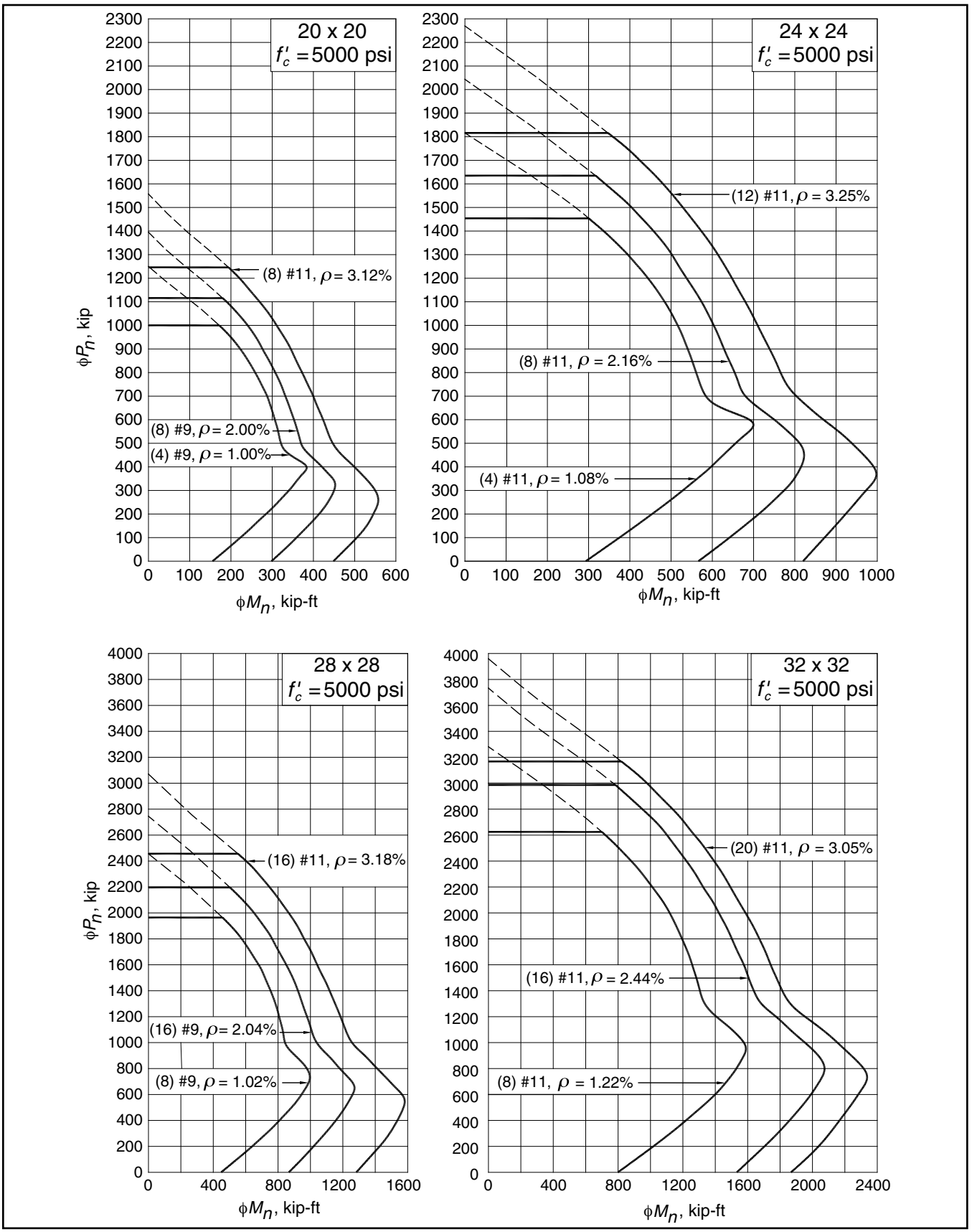
ϕP_n = design axial strength
ϕM_n = design flexural strength
ϕP_O = design axial strength at zero eccentricity
A_g = gross area of the column
δ = moment magnifier (ACI 318-05, Section 10.11–10.13)

The interaction curves have been smoothed for plotting purposes. Exact calculated values may be slightly different.



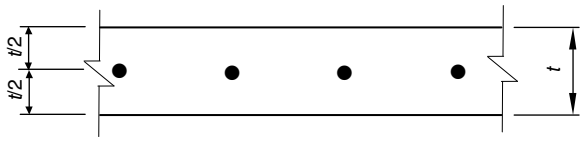


Design Aid 3.12.2 Design Strength Interaction Curves for Precast, Reinforced Concrete Columns (cont.)



3

Design Aid 3.12.5 Partial Interaction Curves for Precast, Prestressed Solid and Sandwich Wall Panels



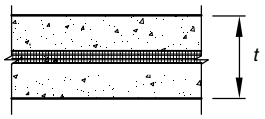
t, in.	Full interaction curve data						
		Partially developed strand force			Fully developed strand force		
	ϕP_o	ϕP_{nb}	ϕM_{nb}	ϕM_c	ϕP_{nb}	ϕM_{nb}	ϕM_c
4	129	46	5.3	0.9	22	4	2.2
6	194	69	11.9	2.0	33	9	5.0
8	259	92	21.2	3.6	45	16	9.0
10	323	115	33	5.6	56	25	14.0

Units for ϕP are kip, units for ϕM are kip-ft.
Values are for 1 ft width.

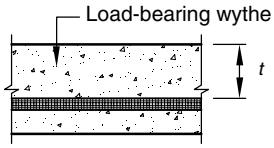
Curves based on minimum prestress of 225 psi normalweight concrete

f'_c = 5000 psi, normalweight

f_{pu} = 270,000 psi

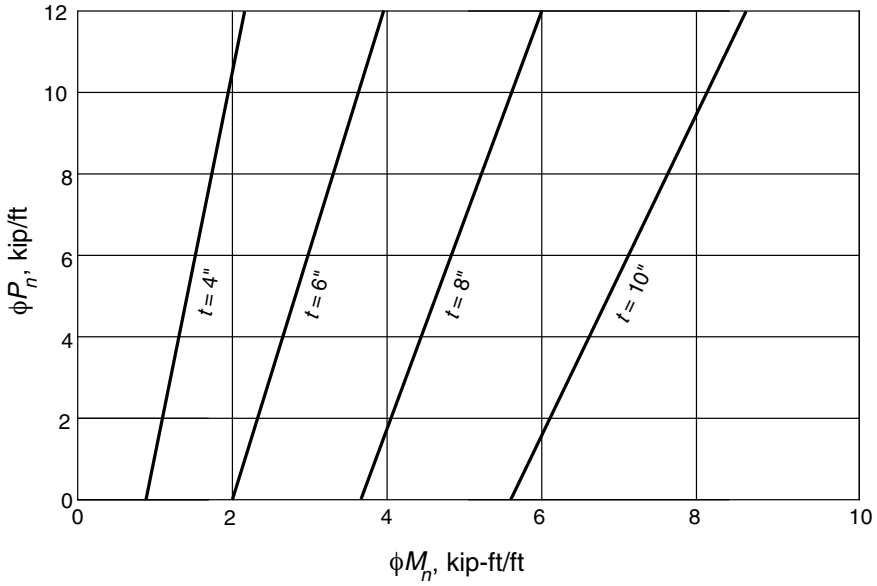


Fully composite sandwich panel

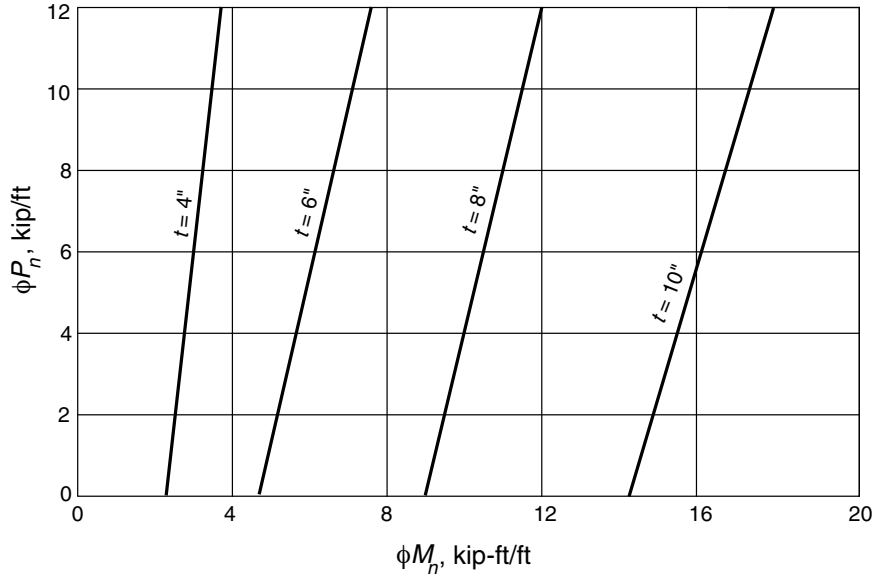


Partially composite or non-composite

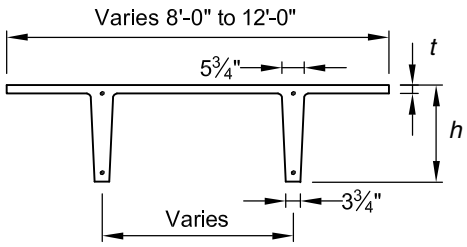
Partially developed strand



Fully developed strand



Design Aid 3.12.3 Design Strength Interaction Curves for Precast, Prestressed Double-Tee Wall Panels



Mark	*h*, in.	*t*, in.	No. strnd	Full interaction curve data (flange in compression)						
					Partially developed strand force			Fully developed strand force		
				ϕP_o	ϕP_{nb}	ϕM_{nb}	ϕM_c	ϕP_{nb}	ϕM_{nb}	ϕM_c
8DT16	16	2	4	864	619	151	45	591	166	95
8DT17	17	3	4	1129	872	178	48	844	195	103
10DT16	16	2	4	997	751	161	45	724	176	96
10DT17	17	3	4	1328	1071	187	48	1043	204	103
12DT16	16	2	4	1130	885	168	46	856	184	97
12DT17	17	3	4	1552	1270	194	48	1242	212	103

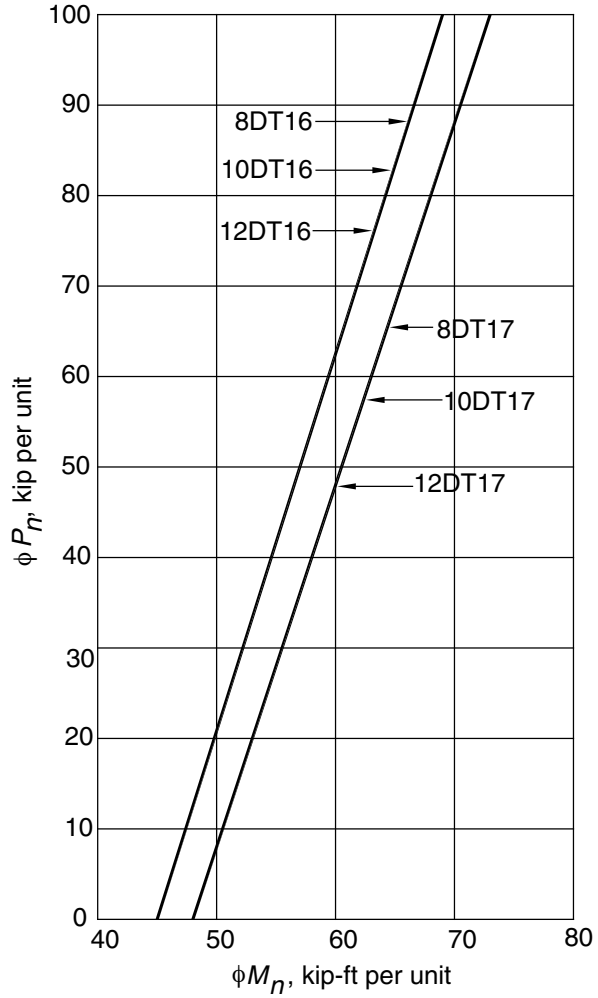
ϕ Varies from 0.75 to 0.65 for partially developed strand.

ϕ Varies from 0.9 to 0.65 for fully developed strand.

Values are applicable for conditions with compression in the flange and tension in the stems.

f_c' = 5000 psi, normalweight
Strand = $^1/_2$ in. diameter
f_{pu} = 270,000 psi

Partially developed strand



Fully developed strand

